

MARION REEF, Australia

WMO: 942980

Lat: 19.096S

Lon: 152.389E

Elev: 25

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	68.2	69.4	N/A	N/A	N/A	N/A	N/A	N/A	31.8	71.7	30.1	72.2	18.4	170	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.4	85.3	N/A	84.5	N/A	83.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.1	0

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(o)	(p)
32.0	29.7	27.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
DB			65.7	88.9	1.8	3.2	64.4	91.2	63.3	93.1	62.3	94.9	60.9	97.2		
WB			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.1	81.1	81.7	80.7	78.8	76.5	74.2	72.7	72.4	73.7	75.7	77.8	80.0
	DBStd	3.63	1.72	1.56	1.66	1.56	1.73	1.87	1.70	1.48	1.64	1.76	1.50	1.68
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9885	965	886	953	863	822	727	703	696	711	795	833	931
	CDD65	4410	500	466	488	413	357	277	238	231	261	330	383	466
	CDH74	31858	5472	5309	5192	3621	2085	743	196	108	385	1366	2786	4595
	CDH80	4371	1168	1363	977	177	30	1	0	0	1	7	73	574
Wind	WSAvg	17.3	16.0	16.0	18.6	19.8	18.7	18.8	18.4	17.1	15.9	16.0	16.1	15.5
Precipitation	PrecAvg	34.5	5.5	6.7	6.5	3.1	2.2	2.0	1.5	1.2	0.6	0.8	1.3	3.1
	PrecMax	55.2	16.5	15.7	20.3	12.8	9.0	7.9	8.5	4.7	1.9	5.7	8.9	12.4
	PrecMin	16.7	0.2	1.3	0.6	0.5	0.2	0.2	0.1	0.1	0.0	0.0	0.1	0.2
	PrecStd	10.1	4.1	4.1	5.5	2.8	2.0	1.7	1.9	1.2	0.5	1.2	1.9	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.6	87.1	85.5	83.7	82.6	79.7	77.6	77.1	78.8	80.6	83.8	85.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	85.2	85.5	84.5	82.3	80.8	78.3	76.5	75.9	77.5	79.2	81.9	83.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	84.3	84.6	83.8	81.6	79.8	77.4	75.7	75.2	76.7	78.5	80.7	83.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	83.6	83.8	83.2	81.0	79.0	76.8	75.1	74.5	75.9	77.9	80.0	82.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.4	3.0	2.7	2.6	2.8	2.6	2.5	2.5	2.6	2.7	2.8
		MCDBR	3.8	4.1	3.8	3.4	3.7	3.1	3.0	2.8	3.1	3.3	4.3	3.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.386	0.391	0.391	0.368	0.354	0.360	0.336	0.340	0.357	0.371	0.376	0.378	
	taud	2.601	2.598	2.586	2.629	2.625	2.599	2.647	2.624	2.549	2.535	2.556	2.595	
	Ebn at Noon	304	299	292	287	279	271	283	292	298	301	305	307	
	Edh at Noon	33	33	32	29	27	27	26	28	33	35	35	33	
All-Sky Solar Radiation	RadAvg	2201	2063	1832	1671	1425	1254	1367	1671	2017	2243	2345	2360	
	RadStd	157	195	174	102	90	94	71	104	65	111	191	166	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page